

Differentiable functions – 1

Derivatives: computation, growth and behaviour at infinity

Homework

1 сентября 2026 года

Theoretical minimum. Leibniz' rule and higher derivatives; derivatives from a power series; the substitution $g = e^{\lambda x} f$ which turns a differential inequality into a monotonicity statement; comparison with the solutions of the corresponding linear equation; l'Hôpital's rule and its hypotheses; the interplay of f , f' and f'' on a half-line.

1. (IMC 2017) Let $f: [0, +\infty) \rightarrow \mathbb{R}$ be continuous with $\lim_{x \rightarrow +\infty} f(x) = L$, finite or infinite. Prove that $\lim_{n \rightarrow \infty} \int_0^1 f(nx) dx = L$.
2. (IMC 2017) Let $f: \mathbb{R} \rightarrow (0, \infty)$ be differentiable and suppose there is $L > 0$ with $|f'(x) - f'(y)| \leq L|x - y|$ for all x, y . Prove that $(f'(x))^2 < 2Lf(x)$ for all x .
3. (IMC 2009) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable with $f(0) = 1$, $f'(0) = 0$ and $f''(x) - 5f'(x) + 6f(x) \geq 0$ for all $x \geq 0$. Prove that $f(x) \geq 3e^{2x} - 2e^{3x}$ for all $x \geq 0$.
4. (IMC 2006) Compare $\tan(\sin x)$ and $\sin(\tan x)$ for $x \in (0, \frac{\pi}{2})$.
5. (IMC 2005) Let $f: (0, \infty) \rightarrow \mathbb{R}$ be twice differentiable with $|f''(x) + 2xf'(x) + (x^2 + 1)f(x)| \leq 1$ for all x . Prove that $\lim_{x \rightarrow \infty} f(x) = 0$.
6. (IMC 2002) Does there exist a continuously differentiable $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) > 0$ and $f'(x) = f(f(x))$ for every x ?
7. (IMC 2012) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuously differentiable with $f'(t) > f(f(t))$ for all t . Prove that $f(f(f(t))) \leq 0$ for all $t \geq 0$.

8. (AKHIMO 2023) Let $f: [0, 1] \rightarrow [0, 1]$ be continuous with a fixed point $\xi \in (0, 1)$. Assume that f is differentiable at ξ and $f'(\xi) < -1$. Prove that there are $0 \leq x_0 < \xi < x_1 \leq 1$ with $f(x_0) = x_1$ and $f(x_1) = x_0$.
9. (Putnam 1998) Let f be a real function on the line with continuous third derivative. Prove that there is a point a with $f(a)f'(a)f''(a)f'''(a) \geq 0$.
10. (Putnam 1999) Let f have a continuous third derivative and suppose f, f', f'', f''' are all positive and $f'''(x) \leq f(x)$ for every x . Show that $f'(x) < 2f(x)$ for all x .